Important Questions

Unit 1:

1. Explain the semantic tags of HTML5 and their importance.
2. What is CSS Flexbox? Explain its properties.
3. Describe the structure of an HTML document with an example.
4. What is CSS? Explain different types of CSS with example.
5. Explain different events of JavaScript.
6. Differentiate between GET and POST methods in AJAX.
7. Explain the HTTP request-response cycle. How does client-server communication work on the web?
8. What are CSS Selectors? Explain different types with examples (element, class, id, pseudo-class).
9. Explain the Box Model in CSS with a diagram. How do margin, padding, border, and content relate?
10. What are JavaScript data types? Explain with examples — primitive vs reference types.
11. Explain the concept of AJAX. How does it work? What are its advantages?

Unit 2:

1. What is NPM? Explain its purpose and significance in NodeJS.
2. What is the Express framework? Discuss its features and advantages.
3. What is NodeJS? Discuss its role and significance in server-side JavaScript development.
4. What is a Web Server? Explain key features.
5. Explain Express JS in brief.
6. What is the event loop in NodeJS? Explain how asynchronous, non-blocking I/O works.
7. Explain async/await in NodeJS with an example. How does it improve over callbacks and promises?
8. What are templating engines in NodeJS? Explain server-side rendering with an example (EJS/Pug).
9. How do you serve static files in ExpressJS? Explain with an example.
10. What is middleware in ExpressJS? Explain with types and examples.

11. Explain routing in ExpressJS. How do you define and handle different HTTP routes?
12. What is the difference between require() and import in NodeJS? Explain module system.
13. Explain routing in ExpressJS with examples of GET, POST, PUT and DELETE routes.
14. What is async/await in JavaScript? How does it differ from Promises and callbacks?

Unit 3:

1. Explain the concept of NoSQL databases. How do they differ from traditional relational databases?
2. Provide steps to connect a NodeJS/ExpressJS application to a MongoDB database.
3. What are cookies and how can they be managed in NodeJS/ExpressJS applications?
4. What is MongoDB? Discuss its architecture and primary features.
5. Write MongoDB shell commands for DML statements.
6. Mention differences between SQL and NoSQL databases.
7. What is a Cookie? How is it handled in Node.js?
8. Explain CRUD operations in MongoDB with shell commands (insertOne, find, updateOne, deleteOne).
9. What is Mongoose? How is it used to define schemas and models in NodeJS?
10. Explain request body parsing in ExpressJS. What is the role of express.json() and express.urlencoded()?
11. What is body-parser middleware in ExpressJS? How is request body parsed?
12. What is user authentication in NodeJS? Explain how to implement login using sessions or JWT.
13. How are SQL databases handled from NodeJS? Explain with an example using MySQL or SQLite.

Important Questions

Unit 4:

1. What is JSX, and why is it used in React applications?
2. What is ReactDOM? Explain in detail.
3. What is LocalStorage in JavaScript? How does it differ from SessionStorage and cookies?
4. Explain the purpose of the Fetch API in React applications.
5. What is composition in React? Give an example.
6. Explain useState with example.
7. Explain useEffect with example.
8. How does ReactDOM render elements into the DOM?
9. What is a component? Explain types with example.
10. Explain the component lifecycle methods in React (mounting, updating, unmounting).
11. What are React Hooks? Explain any two with examples.
12. What are props in React? How are they different from state? Explain with an example.
13. Explain the React component lifecycle methods (mounting, updating, unmounting) in class components.
14. What is Lifting State Up in React? Explain with an example of passing state between components.
15. Explain state management in React. What is "lifting state up"?
16. Explain Inheritance vs Composition in React with examples.
17. What is Composition vs Inheritance in React? Why does React prefer composition?

Important Questions

Unit 5:

1. Explain the concepts of horizontal scaling and vertical scaling with examples.
2. Explain Docker in detail.
3. Explain Kubernetes in detail.
4. Explain Virtual Private Cloud (VPC) in detail.
5. What is Kubernetes? Explain its architecture.
6. What is EC2 in Cloud?
7. What is scaling in cloud? Explain types.
8. What is a Docker container? Why use Docker?
9. What is Cloud Computing? Explain its service models — IaaS, PaaS, and SaaS with examples.
10. Explain the architecture of Kubernetes in detail. Describe master node, worker node, pods, and services.
11. What is a Docker image? How is it different from a Docker container? Explain Dockerfile.
12. What is a Virtual Machine (VM)? How does it differ from a Docker container?
13. Explain the concept of Virtual Private Cloud (VPC). What are its components and benefits?
14. Explain the Dockerfile and Docker image concepts. How do you build and run a Docker image?
15. What is Load Balancing in cloud? How does Kubernetes handle it?
16. What is Ethernet and switching in cloud networking? Explain briefly.
17. Explain the concept of containerization. How is it different from virtualization?